Knowledge Representation

Lecture 2: Actions with Ramifications

dr Anna Maria Radzikowska

Warsaw University of Technology
Faculty of Mathematics and Information Science

E-mail: Anna.Radzikowska@pw.edu.pl

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Outline

1. Dynamic systems with actions' ramifications
2. Language $\mathcal{AR}$ – Action Language with Ramifications
   - Statements in $\mathcal{AR}$
   - Representations in $\mathcal{AR}$
   - Semantics of $\mathcal{AR}$
   - Example

Ramification problem

Ramification problem (Finger, 1987)

Concerns the problem of concisely representing indirect effects of actions (propagation of changes). It is usually unreasonable to explicity enumerate all of the consequences of actions.

Example: Shooting action

Assume that a turkey is walking. Then, as a consequence of a shooting action, it is not walking anymore. Thus *not walking* is a side effect of a shooting action.

Class $\mathcal{AR}$ of dynamic systems

Class of dynamic systems with ramifications

Basic assumptions

- Inertia law.
- Complete information about all actions and all fluents.
- Nondeterminism is allowed.
- Side effects (ramifications) of actions are admitted.
- Action can be unexecutable is some states.
- If an action precondition does not hold, the execution of the action has no effect.

Language $\mathcal{AR}$

To represent this class of dynamic systems we will use action languages of the class $\mathcal{AR}$.

Action language $\mathcal{AR}$: Syntax

Action Language $\mathcal{AR}$: Syntax

Preliminaries

- ***Signature:*** $\Upsilon = (\mathcal{F}, \mathcal{A}c)$, $\mathcal{F}_I \subseteq \mathcal{F}$ (***inertial fluents***).
- ***Formula:*** any propositional statement:

$$\alpha ::= f|\neg\alpha|\alpha \wedge \beta|\alpha \vee \beta|\alpha \rightarrow \beta|\alpha \leftrightarrow \beta.$$

Two specific formulas: $\top$ (truth) and $\perp$ (falsity).

Statements in $\mathcal{AR}$

Value statements

α ***after*** $\mathrm{A}_1, \ldots, \mathrm{A}_n$

observable α ***after*** $\mathrm{A}_1, \ldots, \mathrm{A}_n$

where α is a formula, $\mathrm{A}_i \in \mathcal{A}c$, $i = 1, \ldots, n$, are actions.

INTUITIVE MEANING:

The 1^{st} statement: the condition α always holds after performing the sequence $\mathrm{A}_1, \ldots, \mathrm{A}_n$ of actions.

The 2^{nd} statement: the condition α sometimes (may) holds after performing the sequence $\mathrm{A}_1, ..., \mathrm{A}_n$ of actions.

If $n = 0$, then we use the abbreviation ***initially*** α.

Statements in $\mathcal{AR}$ (cont.)

Effect statements

$$\mathrm{A}\ \textbf{\textit{causes}}\ \alpha\ \textbf{\textit{if}}\ \pi$$

where α, π are formulas and $\mathrm{A} \in \mathcal{A}c$.

INTUITIVE MEANING:

Performing the action A in any state satisfying π leads to a state where α holds.

ABBREVIATIONS:

- If $\pi \Leftrightarrow \top$: A ***causes*** α.
- If $\alpha \Leftrightarrow \bot$: ***impossible*** A ***if*** π.

Statements in $\mathcal{AR}$ (cont.)

Release statement

$$\mathrm{A}\ \textbf{\textit{releases}}\ f\ \textbf{\textit{if}}\ \pi$$

where $\mathrm{A} \in \mathcal{A}c$, $f \in \mathcal{F}_I$, and π is a formula.

INTUITIVELY:

Performing the action A in any state satisfying π might, but need not, change the value of the inertial fluent f.

ABBREVIATION:

if $\pi \Leftrightarrow \top$, we write A ***releases*** f.

E.g., TOSS ***releases*** *heads*.

Statements in $\mathcal{AR}$ (cont.)

Constraint statement

always α

where α is a formula.

INTUITIVELY:
The condition α holds in every state.

E.g., ***always*** $walking \rightarrow alive$.

Statements in $\mathcal{AR}$ (cont.)

Fluent specification statement

noninertial f

where $f \notin \mathcal{F}_I$.

INTUITIVELY:

A fluent f in noninertial whenever it is out of minimization of changes.

Action domain

A finite set of statements of $\mathcal{AR}$ is called an ***action domain***.

Modification of Yale Shooting Problem

Example 2.1

There is a shooter Bill and a turkey Fred. Bill can load the gun, which makes it loaded. He can also shoot the gun which makes the gun unloaded and, in addition, makes Fred dead id the gun was loaded. Initially, the gun was unloaded and Fred is walking. Clearly, whenever Fred is walking, it is alive.

REPRESENTATION IN $\mathcal{AR}$:

initially $\neg loaded \wedge walking$;
always $walking \to alive$;
LOAD ***causes*** $loaded$;
SHOOT ***causes*** $\neg loaded$;
SHOOT ***causes*** $\neg alive$ ***if*** $loaded$.

Russian Turkey Scenario

Example 2.2

There is a shooter Bill and a turkey Fred. Bill can load the gun, which makes it loaded. Bill can also spin the gun, which randomly makes it unloaded. Shooting the gun makes it unloaded and, if the gun was loaded, makes Fred dead. Initially the gun was unloaded and Fred was alive.

Representation in $\mathcal{AR}$:

$$
\begin{array}{l}
\textbf{\textit{initially}}\ \neg loaded \wedge alive; \\
\textsc{Load}\ \textbf{\textit{causes}}\ loaded; \\
\textsc{Shoot}\ \textbf{\textit{causes}}\ \neg loaded; \\
\textsc{Shoot}\ \textbf{\textit{causes}}\ \neg alive\ \textbf{\textit{if}}\ loaded; \\
\textsc{Spin}\ \textbf{\textit{releases}}\ loaded\ \textbf{\textit{if}}\ loaded.
\end{array}
$$

Opening the door

Example 2.3

Inserting a card makes the door open. If one has no card, it is not possible to insert it. Initially the door are closed.

Representation in $\mathcal{AR}$:

$$
\begin{array}{l}
\textbf{\textit{initially}}\ \neg open;\\
\textsc{InsertCard}\ \textbf{\textit{causes}}\ open;\\
\textbf{\textit{impossible}}\ \textsc{InsertCard}\ \textbf{\textit{if}}\ \neg hasCard.
\end{array}
$$

Two switches

Example 2.4

There are two switches, which can be in the position on or off. If both switches are in the same position, the light is on, otherwise it is off. Pressing a switch makes it changed its position.

REPRESENTATION IN $\mathcal{AR}$:

$$
\begin{array}{l}
\textbf{\textit{noninertial}}\ light; \\
\textbf{\textit{always}}\ light \leftrightarrow (switch_1 \leftrightarrow switch_2); \\
\text{TOGGLE1}\ \textbf{\textit{causes}}\ switch_1\ \textbf{\textit{if}}\ \neg switch_1; \\
\text{TOGGLE1}\ \textbf{\textit{causes}}\ \neg switch_1\ \textbf{\textit{if}}\ switch_1; \\
\text{TOGGLE2}\ \textbf{\textit{causes}}\ switch_2\ \textbf{\textit{if}}\ \neg switch_2; \\
\text{TOGGLE2}\ \textbf{\textit{causes}}\ \neg switch_2\ \textbf{\textit{if}}\ switch_2.
\end{array}
$$

Action language $\mathcal{AR}$: Semantics

Semantics of $\mathcal{AR}$

Structure

A ***structure*** for a language $\mathcal{L}$ of the class $\mathcal{AR}$ is a triple $S = (\Sigma, \sigma_0, Res)$ where

- $\Sigma \neq \emptyset$ is a set of states,
- $\sigma_0 \in \Sigma$ is the initial state,
- $Res : \mathcal{A}c \times \Sigma \to 2^{\Sigma}$ is a transition function.

Intuitively, for any action $A \in \mathcal{A}c$ and for any state $\sigma \in \Sigma$, Res assigns all states accessible from σ by performing A.

Moreover,

- $Res(A, \sigma)$ are all such states, which are "as close as possible" to σ;
- Σ is the set of states satisfying all constraints.

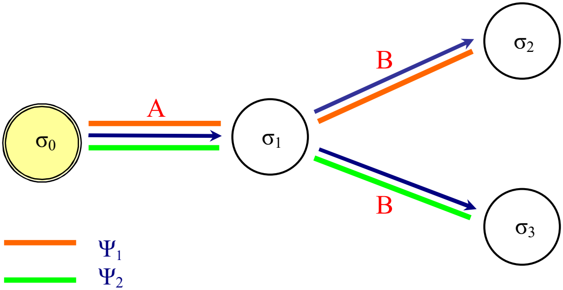
Semantics of $\mathcal{AR}$ (cont.)

Transition function

Let $S = (\Sigma, \sigma_0, Res)$ be a structure. Define a partial mapping $\Psi_S : \mathcal{A}c^* \times \Sigma \rightarrow \Sigma$ as follows

- $\Psi_S(\varepsilon, \sigma) = \sigma$
- if $\Psi_S((\mathrm{A}_1, \ldots, \mathrm{A}_n), \sigma)$, $n \geqslant 1$, is defined, then $\Psi_S((\mathrm{A}_1, \ldots, \mathrm{A}_n), \sigma) \in Res(\mathrm{A}_n, \Psi_S((\mathrm{A}_1, \ldots, \mathrm{A}_{n-1}), \sigma))$

Mapping Ψ depicted



Semantics of $\mathcal{AR}$ (cont.)

Let D be an action domain and let $S = (\Sigma, \sigma_0, Res)$ be a structure for $\mathcal{AR}$.

Inertial fluent

A fluent f is ***inertial*** iff (***noninertial*** f) $\notin D$.

State for D

A state $\sigma : \mathcal{F} \rightarrow \{0, 1\}$ is a ***state for*** D iff for every constraint (***always*** α) $\in D$, it holds $\sigma \models \alpha$.

Satisfiability of value statements

- A value statement (α ***after*** $\mathrm{A}_1, \ldots, \mathrm{A}_n$) is ***true*** in S iff $\Psi_S((\mathrm{A}_1, \ldots, \mathrm{A}_n), \sigma_0) \models \alpha$ for **every** mapping Ψ_S defined before.
- An observation statement (***observable*** α ***after*** $\mathrm{A}_1, \ldots, \mathrm{A}_n$) is ***true*** in S iff $\Psi_S((\mathrm{A}_1, \ldots, \mathrm{A}_n), \sigma_0) \models \alpha$ for **some** mapping Ψ_S defined as before.

Semantics of $\mathcal{AR}$ – functions Res and *New*

Let D be an action domain and let $S = (\Sigma, \sigma_0, Res)$ be a structure for a language $\mathcal{L}$ of the class $\mathcal{AR}$.

Auxiliary function Res_0

Define the auxiliary function $Res_0 : \mathcal{A}c \times \Sigma \rightarrow 2^\Sigma$ as follows: for every $\mathrm{A} \in \mathcal{A}c$ and for every $\sigma \in \Sigma$,

$$Res_0(\mathrm{A}, \sigma) = \{\sigma' \in \Sigma : (A \textbf{\textit{ causes }} \alpha \textbf{\textit{ if }} \pi) \in D \;\&\; (\sigma \models \pi) \Longrightarrow (\sigma' \models \alpha)\}.$$

Function *New*

For every $\sigma \in \Sigma$, for every $\mathrm{A} \in \mathcal{A}c$, and for every $\sigma' \in Res_0(A, \sigma)$, $New(\mathrm{A}, \sigma, \sigma')$ is the set of literals $\overline{f}$ such that $\sigma' \models \overline{f}$ and

1. $f \in \mathcal{F}_I$ and $\sigma(f) \neq \sigma'(f)$, or
2. there is a statement $(\mathrm{A} \textbf{\textit{ releases }} f \textbf{\textit{ if }} \pi) \in D$ and $\sigma \models \pi$.

Model

Let D be an action domain in a language $\mathcal{L}$ of the class $\mathcal{AR}$ and let $S = (\Sigma, \sigma_0, Res)$ be a structure for $\mathcal{L}$. We say that S is a ***model*** of D iff

(M1) Σ is the set of all states for D;

(M2) every value statement and every observation statement in D is true in S;

(M3) for every $\mathrm{A} \in \mathcal{A}c$ and for every $\sigma \in \Sigma$, $Res(\mathrm{A}, \sigma)$ is the set of all states $\sigma' \in \Sigma$ fo which the sets $New(\mathrm{A}, \sigma, \sigma')$ are minimal wrt set inclusion.

Example: YSP

Example 2.1 (cont.)

Recall the YSP:

$$\begin{array}{l}\textit{\textbf{initially}}\ \neg loaded \wedge walking;\\ \textit{\textbf{always}}\ walking \rightarrow alive;\\ \textsc{Load}\ \textit{\textbf{causes}}\ loaded;\\ \textsc{Shoot}\ \textit{\textbf{causes}}\ \neg loaded;\\ \textsc{Shoot}\ \textit{\textbf{causes}}\ \neg alive\ \textit{\textbf{if}}\ loaded.\end{array}$$

Here we have $\Sigma = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5\}$ where

$$\begin{array}{ll}\sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\}\\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\}\\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}.\end{array}$$

Example: YSP (cont.)

Example 2.1 (cont.)

$$\begin{array}{ll} \sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}. \end{array}$$

$$Res_0(\textsc{Load}, \sigma_0) = Res_0(\textsc{Load}, \sigma_1) = Res_0(\textsc{Load}, \sigma_2) = \{\sigma_3, \sigma_4, \sigma_5\}.$$

$$\begin{array}{ll} * & New(\textsc{Load}, \sigma_0, \sigma_3) = \{l\} \\ & New(\textsc{Load}, \sigma_0, \sigma_4) = \{l, \neg w\} \\ & New(\textsc{Load}, \sigma_0, \sigma_5) = \{\neg a, l, \neg w\} \end{array}$$

$$Res(\textsc{Load}, \sigma_0) = \{\sigma_3\}.$$

Example: YSP (cont.)

Example 2.1 (cont.)

$$\begin{array}{ll} \sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}. \end{array}$$

$Res_0(\text{LOAD}, \sigma_1) = \{\sigma_3, \sigma_4, \sigma_5\}$.

$$\begin{array}{rl} & New(\text{LOAD}, \sigma_1, \sigma_3) = \{l, w\} \\ * & New(\text{LOAD}, \sigma_1, \sigma_4) = \{l\} \\ & New(\text{LOAD}, \sigma_1, \sigma_5) = \{\neg a, l\} \end{array}$$

$$Res(\text{LOAD}, \sigma_1) = \{\sigma_4\}.$$

Example: YSP (cont.)

Example 2.1 (cont.)

$$\begin{array}{ll} \sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}. \end{array}$$

$Res_0(\textsc{Load}, \sigma_2) = \{\sigma_3, \sigma_4, \sigma_5\}$.

$$\begin{array}{ll} & New(\textsc{Load}, \sigma_2, \sigma_3) = \{a, l, w\} \\ & New(\textsc{Load}, \sigma_2, \sigma_4) = \{a, l\} \\ * & New(\textsc{Load}, \sigma_2, \sigma_5) = \{l\} \end{array}$$

$$Res(\textsc{Load}, \sigma_2) = \{\sigma_5\}.$$

Example: YSP (cont.)

Example 2.1 (cont.)

$$\begin{array}{ll} \sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}. \end{array}$$

$$Res_0(\textsc{Load}, \sigma_3) = \{\sigma_3, \sigma_4, \sigma_5\}$$

$$\begin{array}{ll} * & New(\textsc{Load}, \sigma_3, \sigma_3) = \emptyset \\ & New(\textsc{Load}, \sigma_3, \sigma_4) = \{\neg w\} \\ & New(\textsc{Load}, \sigma_3, \sigma_5) = \{\neg a, \neg w\} \end{array}$$

$$Res(\textsc{Load}, \sigma_3) = \{\sigma_3\}.$$

Example: YSP (cont.)

Example 2.1 (cont.)

$$\begin{array}{ll} \sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}. \end{array}$$

Similarly,

$$Res_0(\textsc{Load}, \sigma_4) = Res_0(\textsc{Load}, \sigma_5) = \{\sigma_3, \sigma_4, \sigma_5\}.$$

Therefore,

$$\begin{array}{l} Res(\textsc{Load}, \sigma_4) = \{\sigma_4\} \\ Res(\textsc{Load}, \sigma_5) = \{\sigma_5\}. \end{array}$$

Example: YSP (cont.)

Example 2.1 (cont.)

$$\begin{array}{ll}\sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}.\end{array}$$

$$Res_0(\textsc{shoot}, \sigma_0) = Res_0(\textsc{Shoot}, \sigma_1) = Res_0(\textsc{Shoot}, \sigma_2) = \{\sigma_0, \sigma_1, \sigma_2\}$$

$$\begin{array}{ll} * & New(\textsc{shoot}, \sigma_0, \sigma_0) = \emptyset \\ & Res_0(\textsc{Shoot}, \sigma_0, \sigma_1) = \{\neg w\} \\ & Res_0(\textsc{Shoot}, \sigma_0, \sigma_2) = \{\neg a, \neg w\} \end{array}$$

$$Res(\textsc{shoot}, \sigma_0) = \{\sigma_0\}.$$

Analogously, we can easily calculate

$$Res(\textsc{shoot}, \sigma_1) = \{\sigma_1\}$$
$$Res(\textsc{shoot}, \sigma_2) = \{\sigma_2\}.$$

Example: YSP (cont.)

Example 2.1 (cont.)

$$\begin{array}{ll} \sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_1 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}. \end{array}$$

$$Res_0(\textsc{Shoot}, \sigma_3) = \{\sigma_2\}.$$

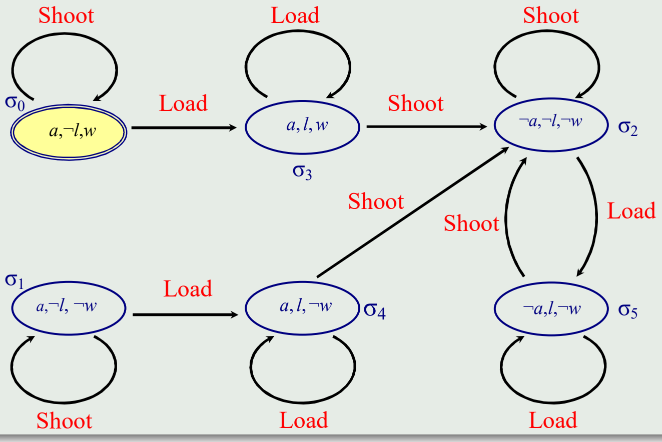
So we immediately get $Res(\textsc{Shoot}, \sigma_3) = \{\sigma_2\}$.
Analogously,

$$Res_0(\textsc{Shoot}, \sigma_4) = Res_0(\textsc{Shoot}, \sigma_5) = \{\sigma_2\},$$

so $Res(\textsc{Shoot}, \sigma_4) = Res(\textsc{Shoot}, \sigma_5) = \{\sigma_2\}$.

Example: YSP (cont.)

Example 2.1 (cont.)



Example: Two switches (cont.)

Example 2.4 (cont.)

Recall

> *There are two switches, which can be in the position on or off. If both switches are in the same position, the light is on, otherwise it is off. Pressing any switch changes its position.*

Representation in $\mathcal{AR}$:

$$
\begin{array}{l}
\textbf{\textit{noninertial}}\ light; \\
\textbf{\textit{initially}}\ switch_1 \land switch_2; \\
\textbf{\textit{always}}\ light \leftrightarrow (switch_1 \leftrightarrow switch_2); \\
\textsc{Toggle1}\ \textbf{\textit{causes}}\ \neg switch_1\ \textbf{\textit{if}}\ switch_1; \\
\textsc{Toggle1}\ \textbf{\textit{causes}}\ switch_1\ \textbf{\textit{if}}\ \neg switch_1; \\
\textsc{Toggle2}\ \textbf{\textit{causes}}\ \neg switch_2\ \textbf{\textit{if}}\ switch_2; \\
\textsc{Toggle2}\ \textbf{\textit{causes}}\ switch_2\ \textbf{\textit{if}}\ \neg switch_2;
\end{array}
$$

Example: Two switches (cont.)

Example 2.4 (cont.)

Here we have $\Sigma = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$ where

$$\begin{array}{ll}
\sigma_0 = \{switch_1, switch_2, light\} & \sigma_2 = \{switch_1, \neg switch_2, \neg light\} \\
\sigma_1 = \{\neg switch_1, \neg switch_2, light\} & \sigma_3 = \{\neg switch_1, switch_2, \neg light\}.
\end{array}$$

Assume that all fluents are inertial. Then

$$Res_0(Toggle1, \sigma_0) = \{\sigma_1, \sigma_3\}$$

$$\begin{array}{ll}
* & New(\textsc{Toggle1}, \sigma_0, \sigma_1) = \{\neg switch_1, \neg switch_2\} \\
* & New(\textsc{Toggle1}, \sigma_0, \sigma_3) = \{\neg switch_1, \neg light\}
\end{array}$$

$$Res(\textsc{Toggle1}, \sigma_0) = \{\sigma_1, \sigma_3\}.$$

So $\textsc{Toggle1}$ in σ_0 is non-deterministic, but the resulting state σ_1 is counterintuitive!

Example: Two switches (cont.)

Example 2.4 (cont.)

$$\begin{array}{ll} \sigma_0 = \{switch_1, switch_2, light\} & \sigma_2 = \{switch_1, \neg switch_2, \neg light\} \\ \sigma_1 = \{\neg switch_1, \neg switch_2, light\} & \sigma_3 = \{\neg switch_1, switch_2, \neg light\}. \end{array}$$

Similarly

$$\begin{array}{ll} & Res_0(\textsc{Toggle}1, \sigma_1) = \{\sigma_0, \sigma_2\} \\ * & New(\textsc{Toggle}1, \sigma_1, \sigma_0) = \{switch_1, switch_2\} \\ * & New(\textsc{Toggle}1, \sigma_1, \sigma_2) = \{switch_1, \neg light\}. \end{array}$$

$$Res(\textsc{Toggle}1, \sigma_1) = \{\sigma_0, \sigma_2\}.$$

Note that only the resulting state σ_2 coincides with our intuition.

Example: Two switches (cont.)

Example 2.4 (cont.)

$$\sigma_0 = \{switch_1, switch_2, light\} \qquad \sigma_2 = \{switch_1, \neg switch_2, \neg light\}$$
$$\sigma_1 = \{\neg switch_1, \neg switch_2, light\} \qquad \sigma_3 = \{\neg switch_1, switch_2, \neg light\}.$$

Now let the fluent *light* be noninertial. Then

$$Res_0(\textsc{Toggle1}, \sigma_0) = \{\sigma_1, \sigma_3\}$$

$$New(\textsc{Toggle1}, \sigma_0, \sigma_1) = \{\neg switch_1, \neg switch_2\}$$
$$* \quad New(\textsc{Toggle1}, \sigma_0, \sigma_3) = \{\neg switch_1\}$$

$$Res(\textsc{Toggle1}, \sigma_0) = \{\sigma_3\}.$$

Although $\sigma_0(light) \neq \sigma_3(light)$, we have $light \notin \mathcal{F}_I$ (noninertial), so $\neg light \notin New(\textsc{Toggle1}, \sigma_0, \sigma_3)$.

Example: Two switches (cont.)

Example 2.4 (cont.)

$$\sigma_0 = \{switch_1, switch_2, light\} \qquad \sigma_2 = \{switch_1, \neg switch_2, \neg light\}$$
$$\sigma_1 = \{\neg switch_1, \neg switch_2, light\} \quad \sigma_3 = \{\neg switch_1, switch_2, \neg light\}.$$

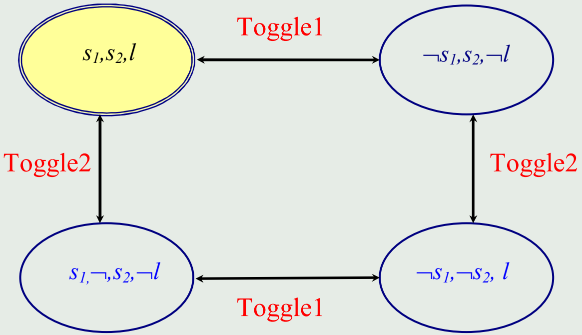
Also

$$Res_0(\textsc{Toggle}1, \sigma_1) = \{\sigma_0, \sigma_2\}$$
$$New(\textsc{Toggle}1, \sigma_1, \sigma_0) = \{switch_1, switch_2\}$$
$$* \quad New(\textsc{Toggle}1, \sigma_1, \sigma_2) = \{switch_1\}$$

$$Res(\textsc{Toggle}1, \sigma_1) = \{switch_2\}.$$

Example: Two switches (cont.)

Example 2.4 (cont.)



Example: Buying a paper

Example 2.5

Consider an action $\textsc{BuyPaper}$ which causes that an agent has a paper A (*HasA*) **or** a paper B (*hasB*).

$$\textsc{BuyPaper}\ \textbf{\textit{causes}}\ hasA \vee hasB.$$

Put

$$\begin{array}{ll} \sigma_0 = \{\neg hasA, \neg hasB\} & \sigma_2 = \{hasA, \neg hasB\} \\ \sigma_1 = \{\neg hasA, hasB\} & \sigma_3 = \{hasA, hasB\}. \end{array}$$

Then

$$\begin{array}{ll} & Res_0(\textsc{BuyPaper}, \sigma_0) = \{\sigma_1, \sigma_2, \sigma_3\} \\ * & New(\textsc{BuyPaper}, \sigma_0, \sigma_1) = \{hasB\} \\ * & New(\textsc{BuyPaper}, \sigma_0, \sigma_2) = \{hasA\} \\ & New(\textsc{BuyPaper}, \sigma_0, \sigma_3) = \{hasA, hasB\} \\ & Res(\textsc{BuyPaper}, \sigma_0) = \{\sigma_1, \sigma_2\}. \end{array}$$

Example: Buying a paper (cont.)

Example 2.5 (cont.)

Problem:

How to obtain the effect of this action such that an agent has both papers?

Possible solution:

$$\textsc{BuyPaper} \; \textit{\textbf{causes}} \; hasA \vee hasB;$$
$$\textsc{BuyPaper} \; \textit{\textbf{releases}} \; hasA \; \textit{\textbf{if}} \; \neg hasA;$$
$$\textsc{BuyPaper} \; \textit{\textbf{releases}} \; hasB \; \textit{\textbf{if}} \; \neg hasB.$$

Example: Buying a paper (cont.)

Example 2.5 (cont.)

$$\sigma_0 = \{\neg hasA, \neg hasB\} \quad \sigma_2 = \{hasA, \neg hasB\}$$
$$\sigma_1 = \{\neg hasA, hasB\} \quad \sigma_3 = \{hasA, hasB\}.$$

So we get:

$$Res_0(\text{BUYPAPER}, \sigma_0) = \{\sigma_1, \sigma_2, \sigma_3\}$$

* $New(\text{BUYPAPER}, \sigma_0, \sigma_1) = \{\neg hasA, hasB\}$
* $New(\text{BUYPAPER}, \sigma_0, \sigma_2) = \{hasA, \neg hasB\}$
* $New(\text{BUYPAPER}, \sigma_0, \sigma_3) = \{hasA, hasB\}$

$$Res(\text{BUYPAPER}, \sigma_0) = \{\sigma_1, \sigma_2, \sigma_3\}.$$

Thank you for your attention!

Questions are welcome.